

Data Flow Diagram

Date	07 July 2026
Team ID	AIML-HDI-2026
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	The end-user who inputs socio-economic variables and receives the HDI forecast.
Rectangle with numbered header	Process (1. GPU Preprocessing, 2. Feature Mapping & Scaling, 3. Hybrid Neural Network)	1. Cleans and imputes the raw dataset on the GPU. 2. Maps dynamic columns and splits/scales features. 3. Executes the dual-branch model to produce the HDI score.
Rectangle (solid fill, no number)	Data Store	The saved trained model (.pt) and Min-Max scaling parameters used at inference time.
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

Figure 3.1 — Data Flow Diagram (Level 1)

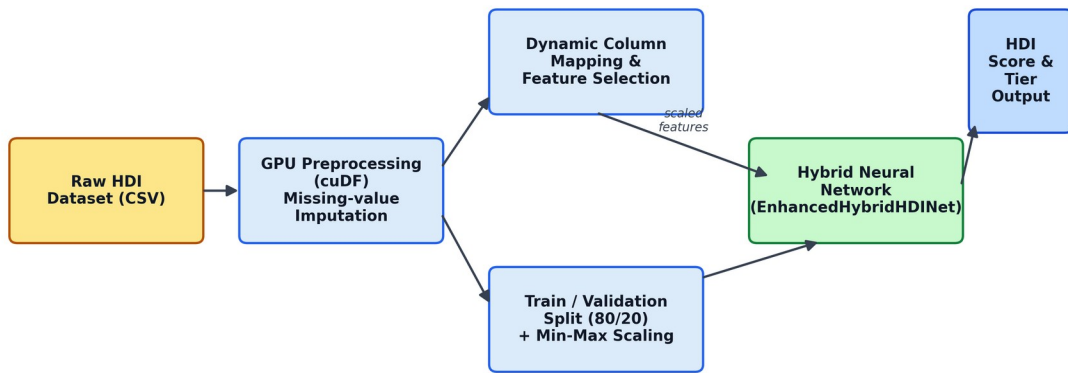


Figure: Data Flow Diagram — Raw dataset → GPU preprocessing → feature mapping & scaling → hybrid neural network → HDI score & tier output.